

For the following information;

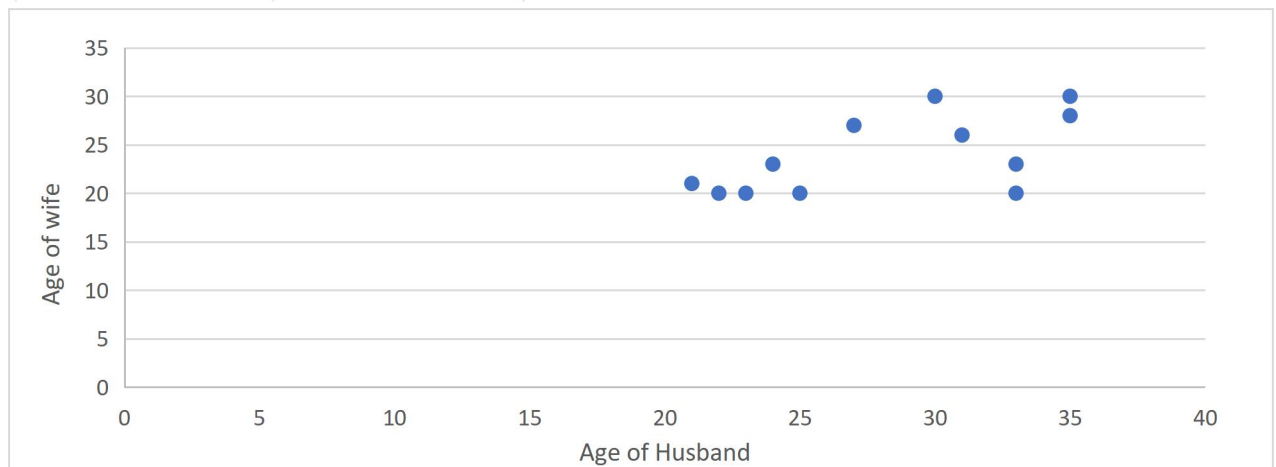
Age of husband	Age of wife	Age of husband	Age of wife
24	23	25	20
22	20	24	23
35	30	34	25
35	28	33	23
23	20	33	27
31	26	30	27
33	23	19	19
33	20	32	19
21	21	21	18
30	30	35	23
27	27	27	22
25	20	19	19

i) Draw scatter diagram.

ii) Interpret the calculated correlation coefficient.

i. Scatter Diagram

Age of husband(x)	Age of wife(y)
24	23
22	20
35	30
35	28
23	20
31	26
33	23
33	20
21	21
30	30
27	27
25	20
339	288
SUM(B18:B29)	SUM(C18:C29)



ii. Interpret the calculated correlation coefficient

	A	B	C	D	E	F	G
11							
12							
13							
14							
15							
16							
17		Age of husband(x)	Age of wife(y)	xy	x^2	y^2	
18		24	23	552	576	529	
19		22	20	440	484	400	
20		35	30	1050	1225	900	
21		35	28	980	1225	784	
22		23	20	460	529	400	
23		31	26	806	961	676	
24		33	23	759	1089	529	
25		33	20	660	1089	400	
26		21	21	441	441	441	
27		30	30	900	900	900	
28		27	27	729	729	729	
29		25	20	500	625	400	
30	Sum	339	288	8277	9873	7088	
31	Formula used	SUM(B18:B29)	SUM(C18:C29)	SUM(D18:D29)	SUM(E18:E29)	SUM(F18:F29)	
32							

50		Symbol	Value	Formula			
51	no. of observations	(n)	12	COUNT(B18:B29)			
52	correlation coeffi	r	0.617495219	((C51*D30-B30*C30)/((SQRT(C51*E30-B30^2))*(SQRT(C51*F30-C30^2))))			
53							

	A	B	C	D	E	
53						
54	Using Excel Tool:					
55						
56		Symbol	Value	Formula		
57	correlation coeff	r	0.617495219	CORREL(B18:B29,C18:C29)		
58						